



How do you efficiently manage and manufacture products for diverse sectors?

We have assigned leaders to each business sector to ensure smooth management and customer support. For example, a leader oversees customer needs in the automotive sector, while another manages the telecommunications segment. These leaders coordinate operations, engage with customers, and ensure process efficiency. Our facility has three automatic production lines—one dedicated to research, development, and small production runs, and the other two for mass production. With nine shifts running daily, we efficiently meet demand. A real-time dashboard tracks operations, and any issues are promptly resolved. Separating production lines prevents conflicts, allowing both R&D and large-scale manufacturing to proceed without delays.

How do you ensure quality across R&D and mass production lines? Do you have two different quality control teams?

Each production line is supported by dedicated line inspectors, NPD, and CFT members to maintain rigorous and specialised quality control. We follow a robust multi-stage quality control process that begins with IQC (incoming quality control), where all incoming materials are thoroughly inspected. During production, IPC (in-process control) monitors each stage to ensure consistency and accuracy. We also employ AOI (automated optical inspection) to detect visual defects, while dedicated PTH and PDS QC teams conduct inspections specific to through-hole components and PCB boards. Finally, all products undergo a comprehensive check by final QC inspectors before dispatch. In total, we conduct quality checks at 8 to 12 critical stages, ensuring high reliability and near-zero rejection rates.

What were the challenges in moving from prototype to mass production?

Many companies design their products using commonly available components, similar to the approach in Korea, China, and Taiwan. They base their designs on market availability and finalise them during the R&D phase. We follow the same strategy, selecting components that are easy to source and have shorter lead times. This approach allows a smooth transition into production without challenges related to component changes. By addressing potential issues during the design stage, we avoid future production obstacles.

What are the challenges in design changes?

We gather customer feedback and execute design and development activities accordingly. When modifications are required in existing designs, we ask customers to specify the changes—whether for the entire design or a specific section, such as power input, frequency, module updates, or communication interfaces. Since multiple segments may be involved, we carefully identify the necessary modifications for a new version (e.g., Version 1.2). Instead of redesigning the entire product, we implement only the required updates. This ensures that any issues related to connectivity, power, or functionality are effectively addressed.

How do you keep equipment working and avoid breakdowns?

We are implementing an AI-driven system within our MES (manufacturing execution system) to enhance operational efficiency. This system connects all manufacturing equipment to a central server, enabling automated monitoring and alerts. For instance, if two preventive maintenance sessions are scheduled within a month, the system sends reminders—typically two days in advance—prompting us to initiate maintenance

activities. This enables timely coordination with machine manufacturers. AI also automates alerts for AMC and maintenance schedules. Additionally, a dedicated team oversees this system, ensuring smooth operations and minimal downtime.

What is the role of AI in modern electronic manufacturing? Are you using it?

We optimise cycle time through AI-driven calculations for component modifications. Our AI module analyses machine cycle times, evaluates the BoM to identify repetitive components on both the top and bottom layers, and adjusts machine operations accordingly. By selecting the most efficient option, we minimise unnecessary loading and unloading, thereby preventing time wastage. This AI-driven strategy reduces changeover time, improves machine efficiency, and ensures a smooth production workflow while maintaining high productivity.

How do you do quality control?

Our quality control follows stringent parameters and adheres to IPC, IATF (for automotive), and ISO standards. We maintain in-house testing facilities that include AI-enabled SMT (surface mount technology), 3D solid-based printing, X-ray inspection, and ICT (in-circuit testing) for feasibility and functional hardware tests. Additionally, our FCT (functional circuit testing) verifies complete box-build assemblies. These in-house labs ensure production stability and significantly reduce field issues. By conducting all tests internally with advanced equipment, we proactively resolve potential challenges and maintain consistent product quality without production disruptions. **EFY**

This interview is an excerpt of a longer discussion. For the full transcript, please visit: <https://efy.co.in/intMay25>

